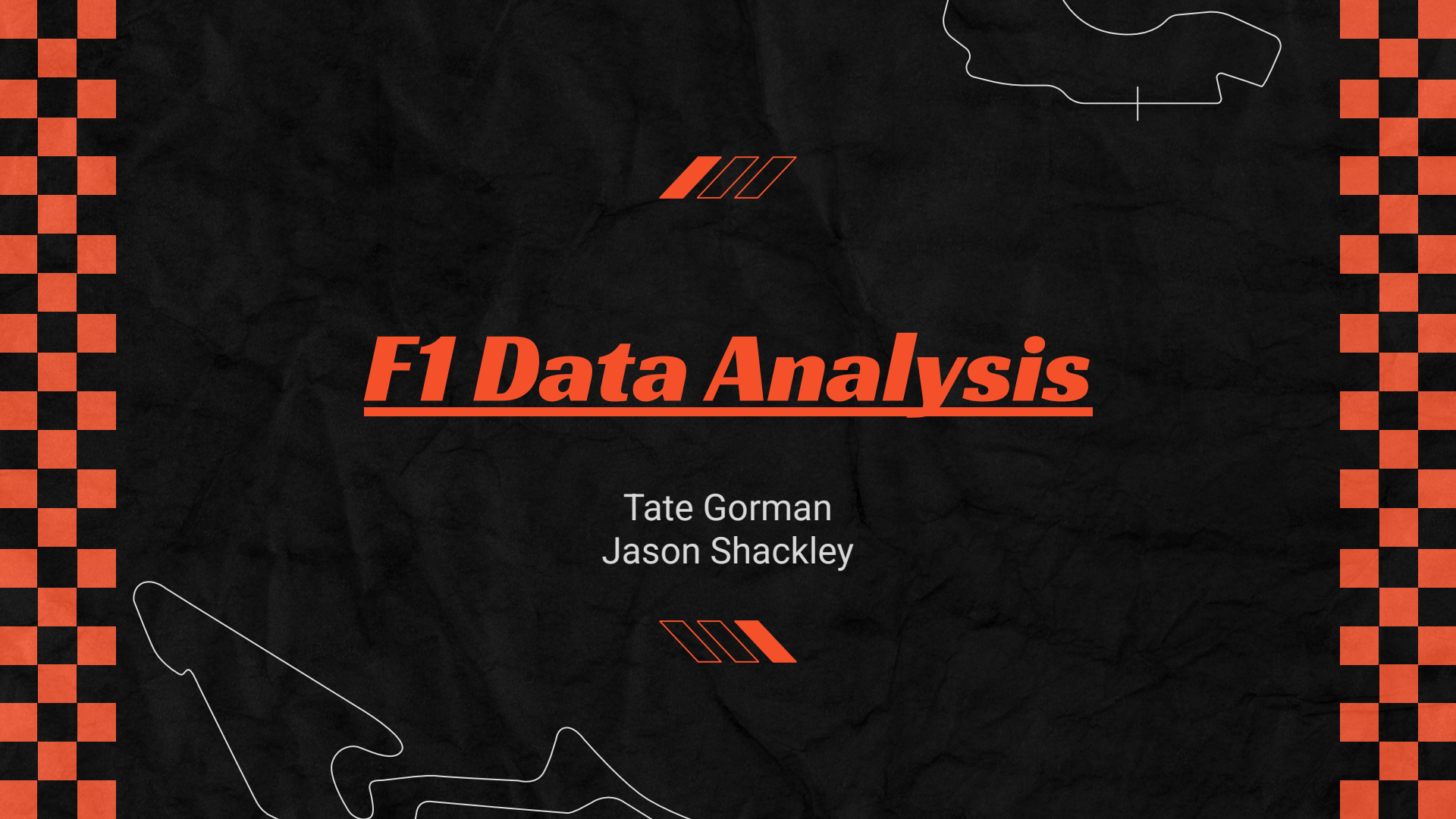
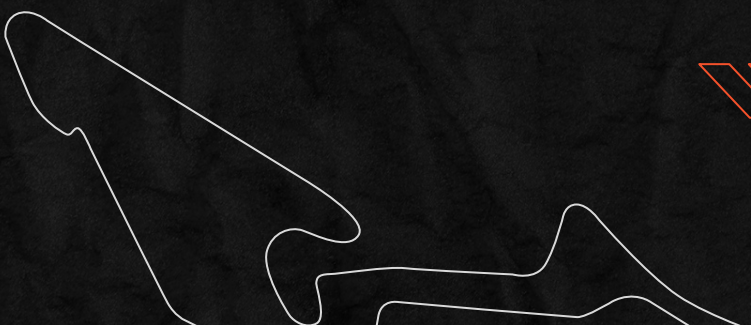




F1 Data Analysis

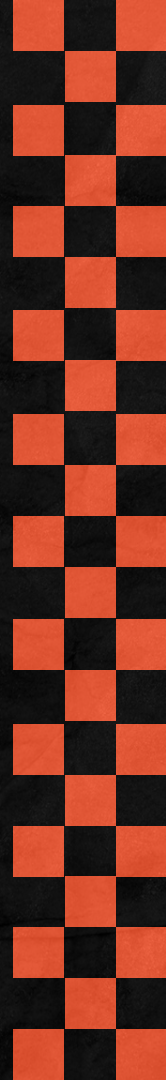
Tate Gorman
Jason Shackley





Why?

- New interest
- Lots of data
- Unpredictability





What determines a driver's success in F1?



Our Data

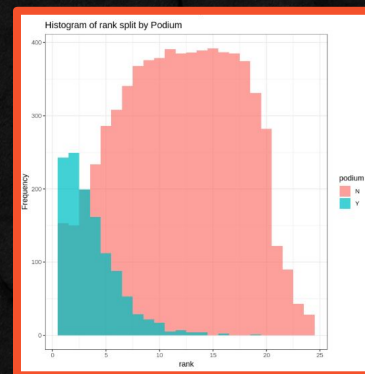
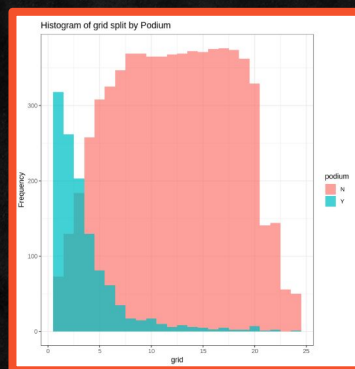
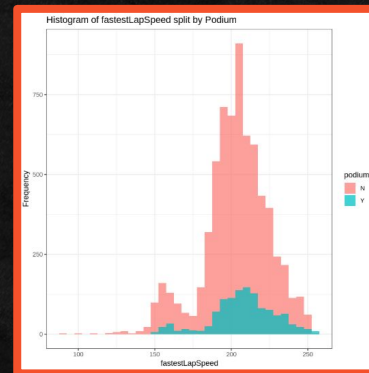
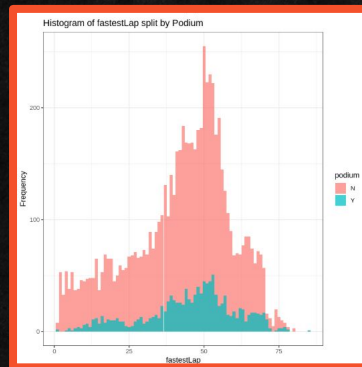
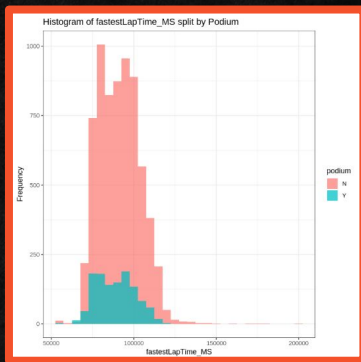
- 2 Datasets from Kaggle
- Race results from many previous F1 races
- Lap times from the 2023 F1 Season



Part 1



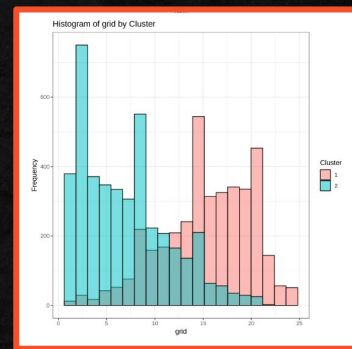
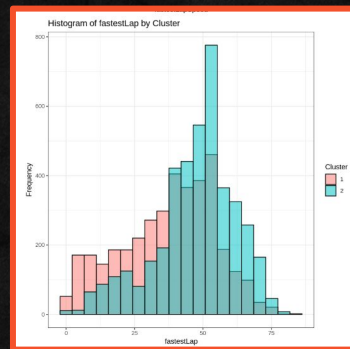
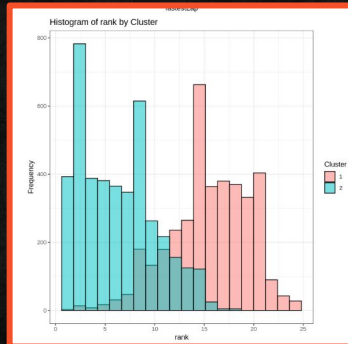
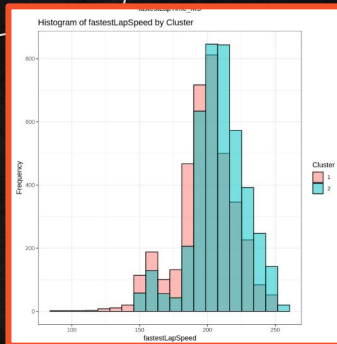
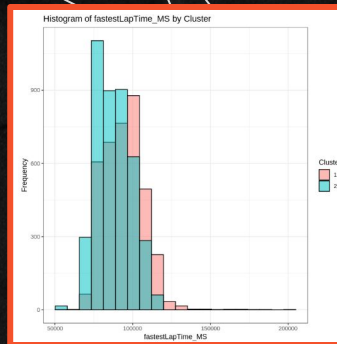
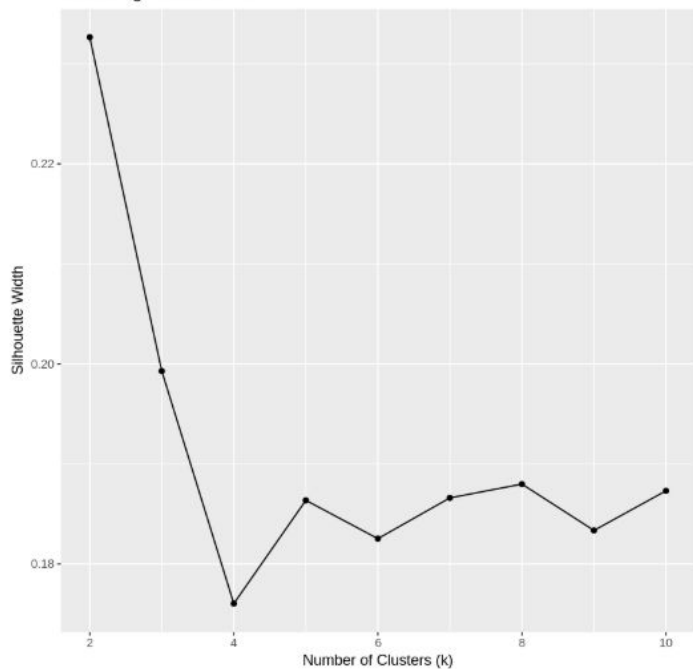
Classification



Clustering

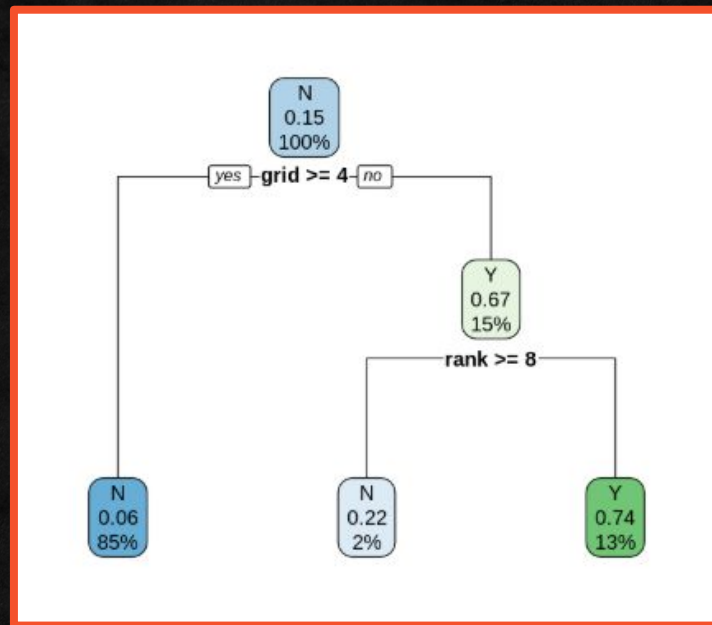
The best k: 2

Clustering Measures vs. k



Decision Tree

- 91% Accuracy
- Classified Yes:
 - Top 4 in grid
 - Top 8 in championship standings



Confusion Matrix

- 142 misclassified cases
- 1454 correctly classified

Confusion Matrix and Statistics

Prediction	Reference	
	N	Y
N	1321	95
Y	47	133

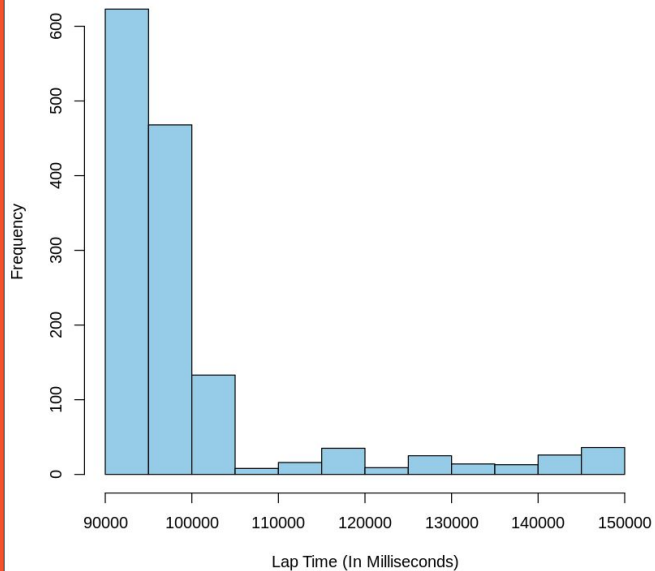


Part 2

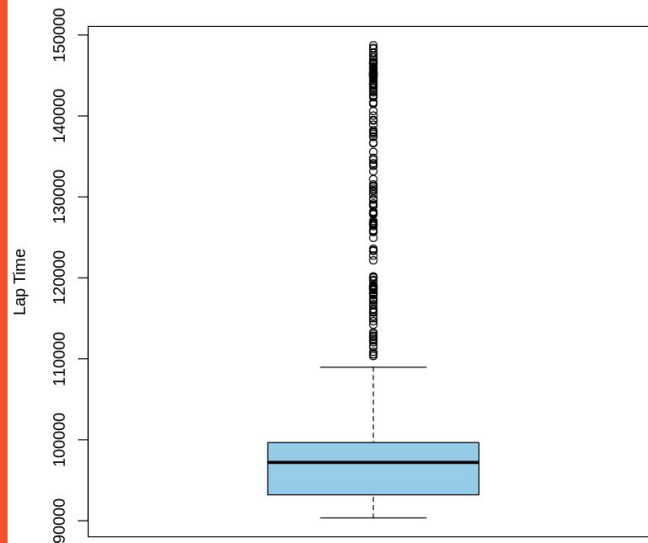


Distribution

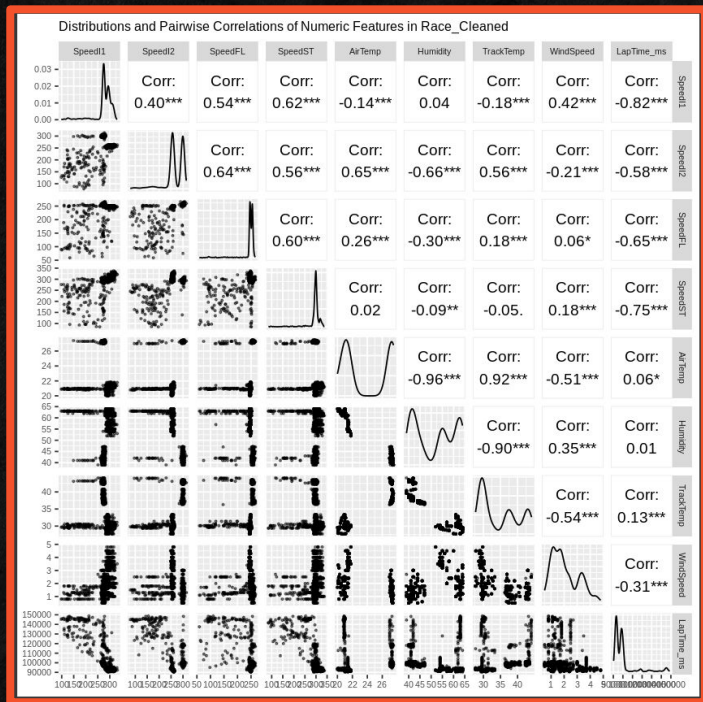
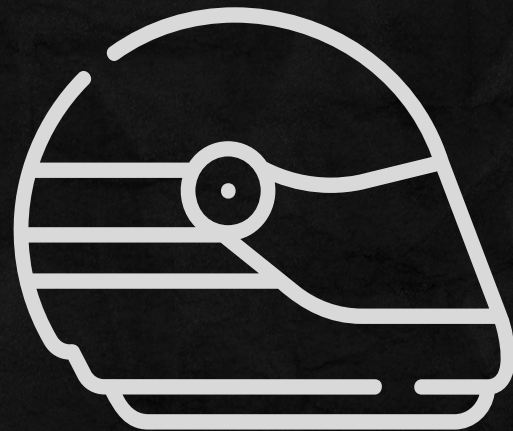
Distribution of Lap Time



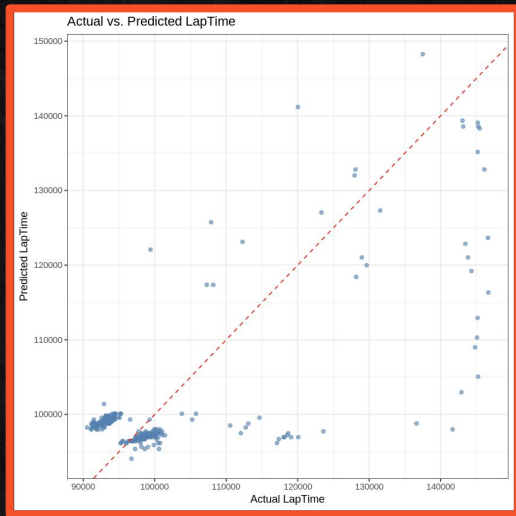
Boxplot of Lap Time



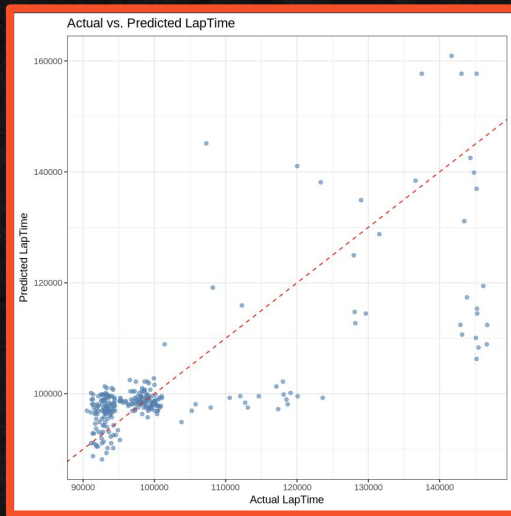
Correlation



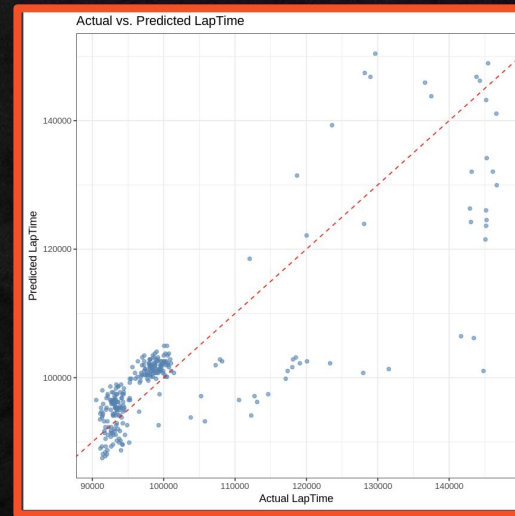
Single Linear Regression



SpeedFL



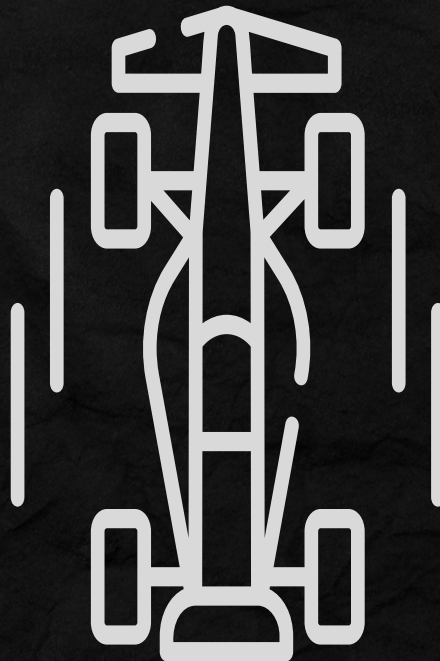
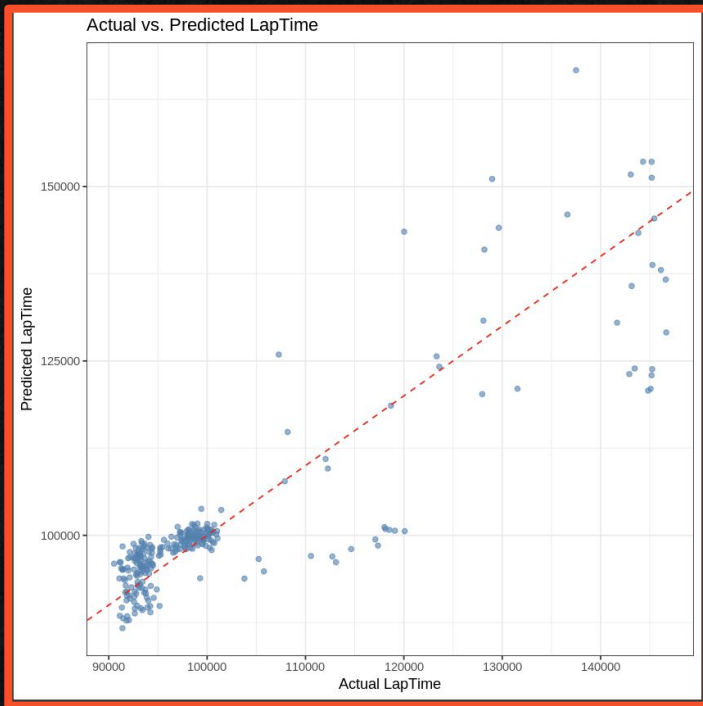
SpeedST



SpeedI1



Multi Linear Regression



Results





Top Classifiers

Grid: Starting race position

Rank: Driver's rank in championship standings prior to race

RMSE VALUES

Mean RMSE- 12794.6465458026

SpeedFL RMSE- 9634.1876809738

SpeedST RMSE- 9247.49437408459

SpeedI1 RMSE- 7914.48349358241

Multi RMSE- 6549.11214993778





Reflection

- ***Strategy***
 - ***Placement***
 - ***Adaptability***
- 



Thank You